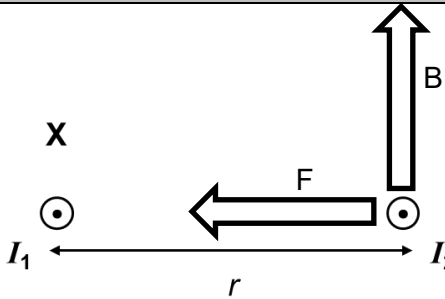
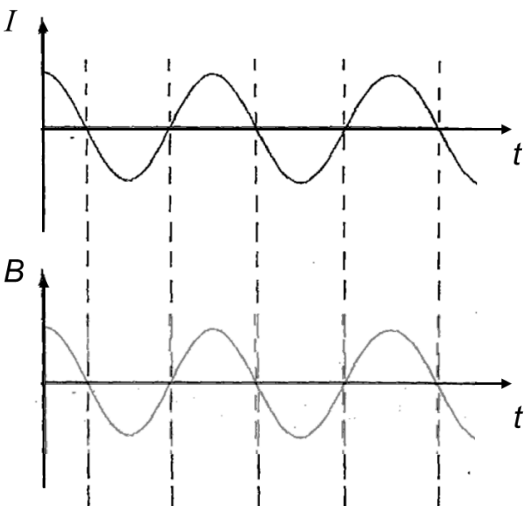
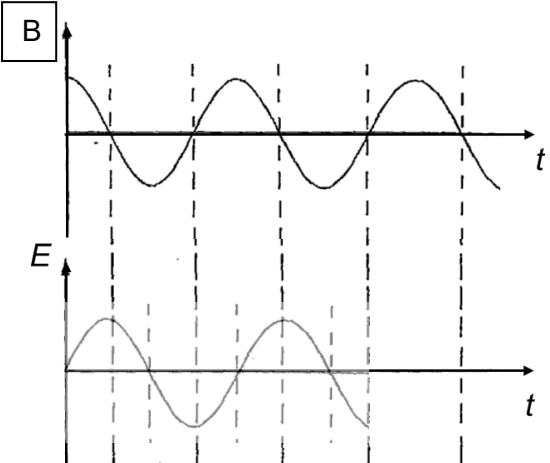


Questions			Answers	Marks
6	(a)	(i) (ii)	 The diagram illustrates the forces between two parallel wires, I_1 and I_2, both carrying current into the page (indicated by dots in circles). I_1 is on the left and I_2 is on the right. A horizontal force vector F points from I_2 towards I_1. A vertical force vector B points upwards from I_2. The distance between the wires is labeled r. An 'X' is marked above I_1.	B1 B1

		(iii)	$F = BIL = \frac{\mu_0 I_1 I_2 L}{2\pi r}$	M1
			$= \frac{\mu_0(5.0)(7.0)}{2\pi(3.0)} = 2.333 \times 10^{-6} \sim 2.3 \times 10^{-6} \text{ N m}^{-1}$	A1
		(b)	Faraday's law of electromagnetic induction states that the e.m.f. induced in a conductor is directly proportional to the rate of change of magnetic flux linkage (or the rate of cutting of magnetic flux)	B1
		(i)	 For solenoid $B = \mu_0 n I \rightarrow B \propto I$	B1
		(ii)	 From Faraday's Law $E = -\frac{d(N\Phi)}{dt} = -\frac{d(NBA \cos \theta)}{dt} \rightarrow E \propto -\frac{dB}{dt}$	B1
		(iii)	$B = \mu_0 n I = (4\pi \times 10^{-7})(10 \times 100)(I) = 1.0 \times 10^{-3}$	M1
			$\rightarrow I = 0.7958 \text{ A} \sim 796 \text{ mA}$	A1

